ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

MECHANICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Mechanical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 B ✓✓ 1.2 A ✓✓ 1.3 A ✓✓ 1.4 A ✓✓ 1.5 A ✓✓ [10]

QUESTION 2

2.1 Any FOUR: wear safety glasses ✓; remove the chuck key before starting ✓; no loose clothing or jewellery, tie back long hair ✓; stop the machine before measuring or cleaning ✓; remove swarf with a brush, never by hand; know where the emergency stop is (4)

2.2 (a) Tightens nuts and bolts to the manufacturer's specified torque ✓ (3)
(b) Finds where compression leaks out of a cylinder (valves, piston rings or head gasket) ✓
(c) Measures the compression pressure in each cylinder to find worn rings or leaking valves ✓

2.3 $12 + 0{,}5 + 0{,}27$ ✓ $= 12{,}77\ \text{mm}$ ✓ (2)

2.4 Any FOUR: test for leaks with soapy water ✓; fit flashback arrestors ✓; never use oil or grease on oxygen fittings ✓; wear goggles with the correct shade of filter lens ✓; open cylinder valves slowly; follow the correct lighting and shut-down order; keep cylinders chained upright (4)

2.5 Acetylene is dissolved in acetone; if the cylinder lies down, acetone can flow out and cause a dangerous flame ✓. An upright, chained cylinder cannot fall and break its valve ✓. (2)

[15]

QUESTION 3

3.1 (a) Hardening: heat to about 30–50 °C above the upper critical temperature, soak, and quench quickly in water or oil ✓; makes steel hard and wear resistant (but brittle) ✓. (8)
(b) Tempering: reheat hardened steel to below the lower critical temperature (about 200–600 °C) and cool ✓; reduces brittleness and increases toughness ✓.
(c) Annealing: heat above the upper critical temperature, soak, and cool very slowly in the furnace ✓; makes steel as soft as possible, relieves stress and improves machinability ✓.
(d) Normalising: heat above the upper critical temperature, soak, and cool in still air ✓; refines the grain structure and relieves internal stresses ✓.

3.2 Giving low-carbon steel a hard, wear-resistant outer skin ✓ while the core stays soft and tough ✓. Any ONE: gears, camshafts, gudgeon pins, crankshaft journals ✓ (3)

3.3 (a) A file slides over hard steel without cutting ✓ but bites easily into soft metal, so it shows how hard the metal is ✓. (4)
(b) Steel rings clearly when struck with a hammer ✓, while cast iron gives a dull sound ✓.

[15]

QUESTION 4

4.1 $A = \pi \times 30^2 \div 4 = 706{,}86\ mm^2$ ✓ (7)
(a) $\sigma = F \div A = 80\ 000 \div 706{,}86$ ✓ $= 113{,}18\ MPa$ ✓
(b) $\varepsilon = \sigma \div E = 113{,}18 \div 210\ 000$ ✓ $= 5{,}39 \times 10^{-4}$ ✓
(c) $\Delta L = \varepsilon \times L = 5{,}39 \times 10^{-4} \times 1\ 500$ ✓ $= 0{,}81\ mm$ shorter ✓

4.2 $UDL = 5 \times 8 = 40\ kN$ acting at 4 m from A ✓ (7)
Moments about A: $\Sigma CWM = \Sigma ACWM$ ✓
$R_B \times 8 = (30 \times 2) + (50 \times 5) + (40 \times 4)$ ✓✓ $= 470\ kN{\cdot}m$
$R_B = 58{,}75\ kN$ ✓
$R_A = (30 + 50 + 40) - 58{,}75$ ✓ $= 61{,}25\ kN$ ✓

4.3 $\Sigma F_x = 120 \cos 30° + 80 \cos 135° + 50 \cos 270°$ ✓ $= 103{,}92 - 56{,}57 + 0 = 47{,}35\ N$ ✓ (8)
$\Sigma F_y = 120 \sin 30° + 80 \sin 135° + 50 \sin 270°$ ✓ $= 60 + 56{,}57 - 50 = 66{,}57\ N$ ✓
$R = \sqrt{(47{,}35^2 + 66{,}57^2)}$ ✓ $= 81{,}69\ N$ ✓
$\theta = \tan^{-1}(66{,}57 \div 47{,}35)$ ✓ $= 54{,}58°$ anticlockwise from the positive x-axis (first quadrant) ✓

4.4 The greatest stress a material can carry ✓ and still return to its original length when the load is removed ✓. Beyond it, the material is permanently deformed ✓. (3)

[25]

QUESTION 5

5.1 (a) $N_C = N_A \times T_A \div T_C$ ✓ $= 1\ 200 \times 24 \div 72$ ✓ $= 400\ r/min$ ✓ (6)
(b) $VR = T_C \div T_A = 72 \div 24 = 3 : 1$ ✓
(c) It makes A and C turn in the same direction ✓ without changing the velocity ratio ✓

5.2 (a) $N_2 = N_1 \times D_1 \div D_2$ ✓ $= 1\ 440 \times 150 \div 400$ ✓ $= 540\ r/min$ ✓ (5)
(b) $v = \pi DN \div 60 = \pi \times 0{,}15 \times 1\ 440 \div 60$ ✓ $= 11{,}31\ m/s$ ✓

5.3 (a) $A_{small} = \pi \times 0{,}025^2 \div 4 = 4{,}91 \times 10^{-4}\ m^2$ ✓ (6)

$p = F \div A = 300 \div 4{,}91 \times 10^{-4}$ ✓ $= 611{,}15\ kPa$ ✓

(b) $A_{large} = \pi \times 0{,}2^2 \div 4 = 0{,}031\ 4\ m^2$ ✓

$F = p \times A = 611\ 155 \times 0{,}031\ 42$ ✓ $= 19\ 200\ N\ (19{,}2\ kN)$ ✓

5.4 Any THREE: the wedge action grips better, so there is less slip ✓; it transmits more power ✓; it runs more quietly ✓; it works on short centre distances; it is less likely to come off the pulley (3)

[20]

QUESTION 6

6.1 Preventive maintenance is planned, regular servicing ✓ done before a breakdown happens, e.g. oil changes ✓. Corrective maintenance is repairing or replacing parts ✓ after a fault or breakdown has happened ✓. (4)

6.2 Any THREE, one mark for the cause and one for prevention: (6)

Porosity: gas trapped from dirty metal or damp electrodes ✓ – clean the metal, use dry electrodes ✓

Undercut: current too high or wrong electrode angle ✓ – lower the current, use the correct angle ✓

Slag inclusion: slag not removed between runs ✓ – chip and brush every run ✓

Cracks: cooling too fast – preheat and cool slowly

Incomplete penetration: current too low or travel too fast – increase current, weld slower

6.3 Any FIVE: spring washer ✓; lock nut ✓; castle nut and split pin ✓; tab washer ✓; nylon-insert (Nyloc) nut ✓; thread-locking compound; locking wire (5)

[15]

TOTAL: 100

END OF MARKING GUIDELINE